

Self-Adaptive Silicon Photonic Equalizer for Polarization-Insensitive and Multi-Wavelength 200-Gbps O-Band Interconnects

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Abstract We demonstrate a self-adaptive silicon photonic equalizer for polarization-insensitive, multi-wavelength division O-band transmission. The equalizer simultaneously equalizes 2x50-Gbaud PAM4 signals transmitted in a WDM system, improving the received sensitivity about 4 dB. ©2025 The Author(s)

Introduction

As artificial intelligence technologies rapidly develop, such as large-scale predictive models like GPT, the global demand for high-performance computing is increasing. Data centers, crucial for processing and storing vast amounts of data, encounter significant challenges in managing data efficiently and cost-effectively. Optical interconnect technologies within data centers must continually improve to meet the growing demands for higher throughput and bandwidth. As data processing needs rise, the performance requirements for optical modules also increase. This results in higher costs and energy consumption [1,2].

In short-distance data center interconnects, signal distortions often occur due to the bandwidth limitations of intensity modulators, photodetectors, and other electronic components, as well as fiber dispersion. These distortions usually require compensation through high-cost,

energy-intensive DSP chips [3]. Optical equalization techniques, in contrast to electronic equalizers used after optical detection, pre-adjust signals before detection. This reduces the complexity and cost of electronic signal processing [4]. Advances in photonic integration have further refined optical equalization. By integrating optical equalizers onto chips, the system compensates for inter-symbol interference caused by dispersion and device bandwidth limitations, enhancing signal transmission [5–8].

The signal quality of optical fiber communication systems in O band is mainly affected by the bandwidth limitation of optoelectronic devices rather than dispersion. As a result, the inter-symbol interference related to dispersion is minimal. We develop a polarization-insensitive adaptive optical equalizer which is controlled by particle swarm optimization (PSO) algorithm for voltage values of phase

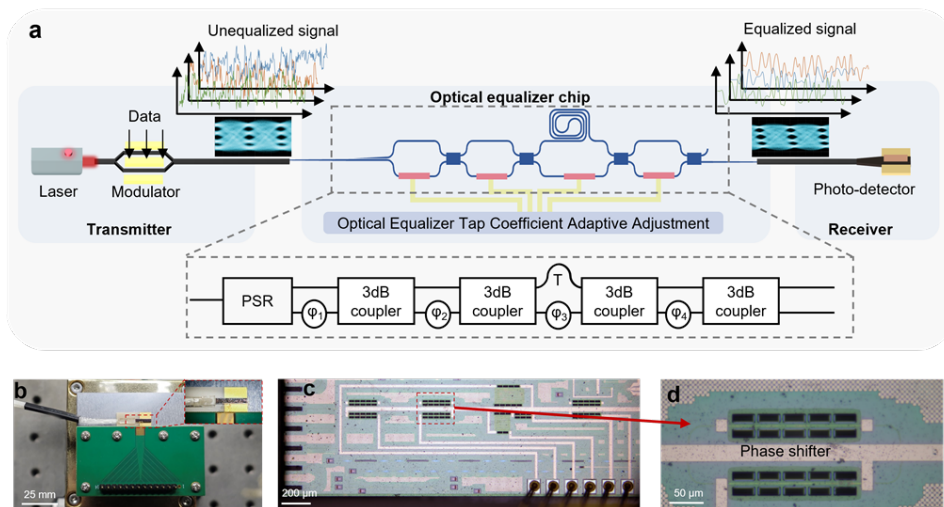


Fig. 1: (a) Schematic of the polarization-insensitive adaptive optical equalizer chip for short-distance optical interconnects, (b) Photograph of the optical equalizer chip, (c) Optical micrograph of the optical equalizer chip, (d) Micrograph of the phase shifter.

shifter. The equalizer uses silicon photonics technology for ultra-low insertion loss on-chip equalization and simultaneously supports optical processing of wavelength division multiplexing (WDM) signals. We demonstrate the equalization performance of the proposed optical equalizer to 2x50-Gbaud PAM4 signals transmitted in O band in a WDM system with limited bandwidth. Optical equalizer reduces inter-symbol interference caused by both electronic and optoelectronic components. The solution offers a cost-effective optical interconnect for data centers.

Operation principle

Figure 1(a) illustrates the schematic of polarization-insensitive optical equalizer chip, designed for short-distance optical interconnects. This chip can process non-equalized signals with any input polarizations in optical domain. The signal, emitted from the laser, is first modulated by a modulator with 25-GHz 3-dB bandwidth. Then optical signals are equalized by the optical equalizer, and finally sent to the photodetector. The chip mainly consists of a polarization rotator splitter (PSR), 3dB couplers, phase shifters, and delay lines. The PSR distributes light with different polarization components into separate paths, which are adjusted by a two-stage mach-zehnder interferometer (MZI) formed by the phase shifters and 3dB couplers, prior to entering the delay lines. After passing through the delay lines, the paths are recombined by another MZI stage enabling amplitude modulation and signal equalization. This effectively forms a double-tap optical finite impulse response (FIR) filter with adaptively adjustment coefficients.

As shown in Fig. 1(b), the optical equalizer chip is mounted on a customized PCB and encapsulated using arrayed fiber technology and

wire bonding. The chip is fabricated on a silicon-on-insulator (SOI) wafer with a 220-nm-thick silicon layer. Figure 1(c) shows the micrograph of the chip, measuring 2300 μ m by 800 μ m. To reduce power consumption further, we employ thermo-optic phase shifters with air trenches, consuming only 1.2mW/ π , as shown in Fig. 1(d). This approach significantly lowers both power consumption and thermal crosstalk. The total power consumption of the chip is less than 6mW, which is substantially lower than that of traditional electronic equalizers.

Experimental setup and results

The experimental setup for validating the performance of optical equalizer is shown in Fig. 2(a). Two external cavity lasers (ECLs) as WDM light sources are combined using a 50:50 coupler, while only one laser is used in the single-wavelength system. The carrier passes through a polarization controller (PC) and is modulated with 50-Gbaud PAM4 signals generated by an arbitrary waveform generator (AWG). The modulated signal enters the optical equalizer chip after adjusting the polarization state by the PC. The output signal is amplified by a praseodymium-doped fiber amplifier (PDFA) and its power is controlled by a variable optical attenuator (VOA) before being sent to a photodetector (PD) for opto-electric conversion. The acquired signal is sampled by an oscilloscope (OSC) and transferred to a computer for analysis, where the PSO algorithm adjusts the on-chip phase shifters to optimize the signal. Figure 2(b) shows that the bit error rate (BER) decreases with each iteration, stabilizing within 10 iterations. After training, we test the BER curves at different received optical power levels, as shown in Fig. 2(c). By varying the input

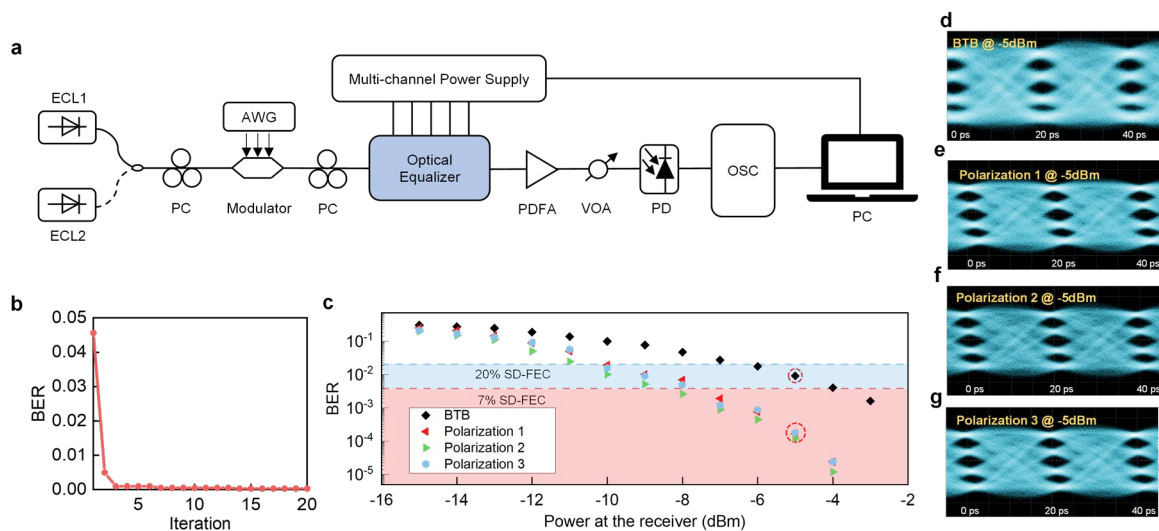


Fig. 2: (a) Optical equalizer test system setup diagram, **(b)** Iteration curve during system training, **(c)** BER curves for the system under BTB and different polarization state inputs, **(d) - (g)** Eye diagrams at a -5 dBm received optical power.

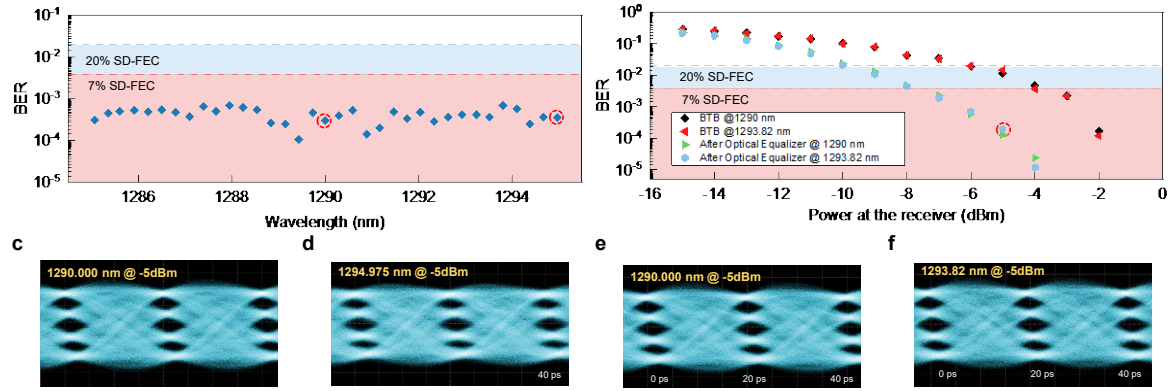


Fig. 3: (a) BER values for different wavelengths at -5 dBm received optical power, (b) EBR curve in the wavelength-division multiplexing (WDM) system, (c)-(d) Eye diagrams for different wavelengths at -5 dBm received optical power, (e)-(f) Eye diagrams for different wavelengths in the WDM system at -5 dBm received optical power

polarization, the system is retrained, and the BER curves for each polarization state are plotted. The results indicate that the equalized system provides a 4dB improvement in received optical power compared to the back-to-back (BTB) system at the same BER. Figures 2(d)-(g) displays the eye diagrams at -5 dBm received optical power for both the BTB system and various polarization states, respectively. The equalized eye diagrams show significant improvement in quality compared to the non-equalized ones.

One of the main advantages of optical equalizers over electronic ones is their ability to simultaneously equalize multiple optical signals under width-limited conditions. This feature makes them ideal for multiplexing systems, reducing DSP-related power consumption. Currently, WDM technology enhances communication capacity in mainstream optical module systems. We have validated the performance of our designed optical equalizer chip within a WDM system.

Initially, the system is trained and optimized at a wavelength of 1290nm. Maintaining the same training configuration, we then adjust the output wavelength of the external cavity laser to validate multiple wavelengths. As depicted in Fig. 3(a), we assess the BER for wavelengths ranging from 1285nm to 1295nm at -5dBm received optical power. The optimized system maintains consistent performance, with all measurements below the 7% overhead SD-FEC threshold. Figures 3(d)-(e) display the eye diagrams for 1290nm and 1294.975nm wavelengths, showing similar results. Figure 3(b) compares the BER curves of the WDM system for two different laser configurations. Both configurations at 1290nm and 1293.82nm demonstrate nearly identical performance, improving by 4dB over the BTB system. The related eye diagrams are illustrated

in Fig. 3 (f)-(g). These experimental findings confirm that the optical equalizer chip excels in the WDM systems.

Conclusion

In conclusion, we present a polarization-insensitive adaptive optical equalizer on a silicon photonics platform, effectively mitigating signal degradation caused by bandwidth limitations in O-band short-distance interconnects. Using a PSO algorithm, the equalizer adaptively adjusts 50-Gbaud PAM4 signals, reducing inter-symbol interference and enhancing transmission performance. The device achieves ultra-low power consumption (<6 mW), significantly lower than conventional electronic equalizers, and supports robust multi-wavelength operation. Experimental validation in WDM systems demonstrates consistent performance, underscoring the equalizer's potential for high-capacity, energy-efficient optical interconnects in data centers. This work addresses current challenges in optical interconnect technology and paves the way for future advancements in scalable, cost-effective communication solutions.

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